C

ORDINARY DIFFERENTIAL EQUATIONS • UNIT 5.01

Variation of Parameters & Greens Kernel

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

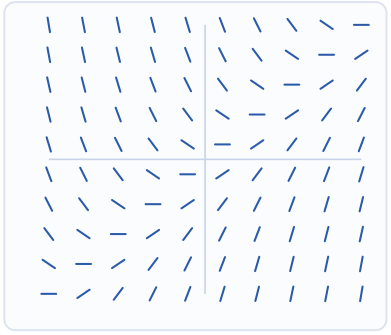
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 B	Q2 C	Q3 A	Q4 D	Q5 B
Q6 A	Q7 C	Q8 D	Q9 C	Q10 D
Q11 B	Q12 A	Q13 D	Q14 C	Q15 B
Q16 A	Q17 D	Q18 B	Q19 B	Q20 A
Q21 B	Q22 D	Q23 C	Q24 C	Q25 C

Variation of Parameters & Greens Kernel Practice Exercises

1 MCQ

Find a particular solution of $y'' - y = 2$ using variation of parameters.



A 2

B -2 ✓

C -1

D $-\frac{1}{2}$

WORKED SOLUTION & VERIFICATION

$y_1 = e^x$, $y_2 = e^{-x}$, $W = -2$. $u_1' = -\frac{y_2(2)}{W} = e^{-x} \Rightarrow u_1 = -e^{-x}$; $u_2' = \frac{y_1(2)}{W} = -e^x \Rightarrow u_2 = -e^x$. So $y_p = u_1 y_1 + u_2 y_2 = -e^{-x}e^x - e^x e^{-x} = -2$.

2 MCQ

Find a particular solution of $y'' - 4y' + 4y = 6$ using variation of parameters.

A $-\frac{3}{2}$

B 3

C $\frac{3}{2}$ ✓

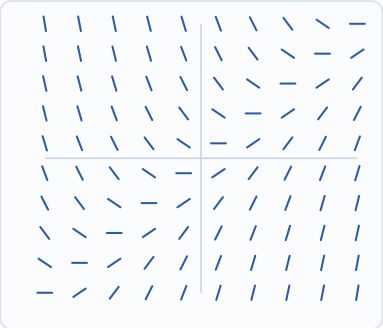
D 6

WORKED SOLUTION & VERIFICATION

$y_1 = e^{2x}$, $y_2 = xe^{2x}$ (repeated root), $W = e^{4x}$. $u_1' = -\frac{y_2(6)}{W} = -6xe^{-2x}$, $u_2' = \frac{y_1(6)}{W} = 6e^{-2x}$; integrating and substituting back gives $y_p = \frac{3}{2}$ (constant) — matching what undetermined coefficients would give directly, $A = 6/4$.

3 MCQ

Find a particular solution of $y'' + 4y = 8$ using variation of parameters.



- **A** 2 ✓
- **B** -2
- **C** 4
- **D** 8

WORKED SOLUTION & VERIFICATION

$y_1 = \cos 2x$, $y_2 = \sin 2x$, $W = 2$. $u_1' = -\dfrac{y_2(8)}{W} = -4\sin 2x \Rightarrow u_1 = 2\cos 2x$; $u_2' = \dfrac{y_1(8)}{W} = 4\cos 2x \Rightarrow u_2 = 2\sin 2x$. So $y_p = u_1 y_1 + u_2 y_2 = 2\cos^2 2x + 2\sin^2 2x = 2$.

4 MCQ

Find a particular solution of $y'' + y = \cot x$ (for $0 < x < \pi$) using variation of parameters.

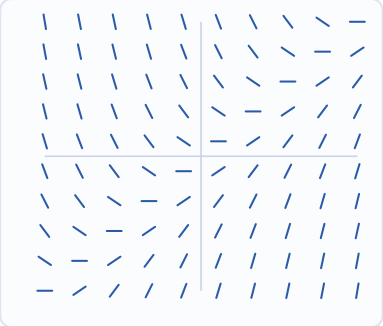
- **A** $\sin x \ln|\csc x + \cot x|$
- **B** $-\cos x \ln|\csc x + \cot x|$
- **C** $-\sin x \ln|\sec x + \tan x|$
- **D** $-\sin x \ln|\csc x + \cot x|$ ✓

WORKED SOLUTION & VERIFICATION

$y_1 = \cos x$, $y_2 = \sin x$, $W = 1$. $u_1' = -\sin x \cot x = -\cos x \Rightarrow u_1 = -\sin x$; $u_2' = \cos x \cot x = \csc x - \sin x \Rightarrow u_2 = \cos x - \ln|\csc x + \cot x|$. So $y_p = u_1 y_1 + u_2 y_2 = -\sin x \cos x + (\cos x - \ln|\csc x + \cot x|)\sin x = -\sin x \ln|\csc x + \cot x|$ (the $\sin x \cos x$ terms cancel).

5 MCQ

Find a particular solution of $y'' - 4y = e^{2x}$ using variation of parameters.



A $(\frac{x}{4} + \frac{1}{16})e^{2x}$

B $(\frac{x}{4} - \frac{1}{16})e^{2x}$ ✓

C $(\frac{x}{2} - \frac{1}{16})e^{2x}$

D $-(\frac{x}{4} - \frac{1}{16})e^{2x}$

WORKED SOLUTION & VERIFICATION

$y_1 = e^{2x}$, $y_2 = e^{-2x}$, $W = -4$. Here $r = 2$ matches a root — no modification is needed for variation of parameters, unlike undetermined coefficients. $u_1' = \frac{1}{4} \Rightarrow u_1 = \frac{x}{4}$; $u_2' = -\frac{e^{4x}}{4} \Rightarrow u_2 = -\frac{e^{4x}}{16}$. So $y_p = u_1 y_1 + u_2 y_2 = (\frac{x}{4} - \frac{1}{16})e^{2x}$.

6 MCQ

Find a particular solution of $y'' + 4y = \csc 2x$ (for $0 < x < \pi/2$) using variation of parameters.

A $-\frac{x}{2}\cos 2x + \frac{1}{4}\sin 2x \ln|\sin 2x|$ ✓

B $\frac{x}{2}\cos 2x + \frac{1}{4}\sin 2x \ln|\sin 2x|$

C $-\frac{x}{2}\cos 2x - \frac{1}{4}\sin 2x \ln|\sin 2x|$

D $-\frac{x}{4}\cos 2x + \frac{1}{4}\sin 2x \ln|\sin 2x|$

WORKED SOLUTION & VERIFICATION

$y_1 = \cos 2x$, $y_2 = \sin 2x$, $W = 2$. $u_1' = -\frac{1}{2} \Rightarrow u_1 = -\frac{x}{2}$; $u_2' = \frac{1}{2}\cot 2x \Rightarrow u_2 = \frac{1}{4}\ln|\sin 2x|$. So $y_p = u_1 y_1 + u_2 y_2 = -\frac{x}{2}\cos 2x + \frac{1}{4}\sin 2x \ln|\sin 2x|$.

7 MCQ

Find a particular solution of $y'' - 6y' + 9y = \frac{e^{3x}}{x^2}$ (for $x > 0$) using variation of parameters.

A $e^{3x}(\ln x + 1)$

B $-e^{3x}(\ln x - 1)$

C $-e^{3x}(\ln x + 1)$ ✓

D $-e^{3x}\ln x$

WORKED SOLUTION & VERIFICATION

$y_1 = e^{3x}$, $y_2 = xe^{3x}$ (repeated root), $W = e^{6x}$. $u_1' = -\frac{1}{x} \Rightarrow u_1 = -\ln x$; $u_2' = \frac{1}{x^2} \Rightarrow u_2 = -\frac{1}{x}$. So $y_p = u_1 y_1 + u_2 y_2 = -e^{3x}\ln x - e^{3x} = -e^{3x}(\ln x + 1)$.

8 MCQ

Find a particular solution of $y'' + 4y' + 3y = e^{-x}$ using variation of parameters.

A $(\frac{x}{2} + \frac{1}{4})e^{-x}$

B $(x - \frac{1}{4})e^{-x}$

C $-(\frac{x}{2} - \frac{1}{4})e^{-x}$

D $(\frac{x}{2} - \frac{1}{4})e^{-x}$ ✓

WORKED SOLUTION & VERIFICATION

$y_1 = e^{-x}$, $y_2 = e^{-3x}$ (roots $-1, -3$), $W = -2e^{-4x}$. $u_1' = \frac{1}{2} \Rightarrow u_1 = \frac{x}{2}$; $u_2' = -\frac{e^{2x}}{2} \Rightarrow u_2 = -\frac{e^{2x}}{4}$. So $y_p = u_1 y_1 + u_2 y_2 = (\frac{x}{2} - \frac{1}{4})e^{-x}$.

9 MCQ

The second-order linear homogeneous ODE $F''(w) - 50F'(w) + 621F(w) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 31

B 73

C **50** ✓

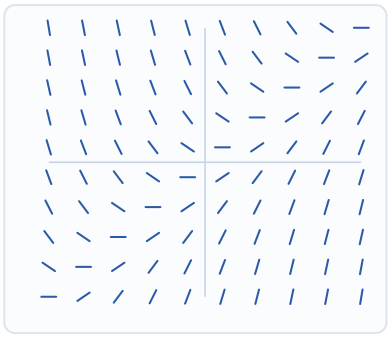
D 53

WORKED SOLUTION & VERIFICATION

From $\lambda^2 - 50\lambda + 621 = 0$, the sum of roots is 50.

10 MCQ

Consider the second-order linear nonhomogeneous differential equation $y'' - y = \sin x$. If $y_p(x) = A\cos x + B\sin x$ is a particular solution, find the values of the constants A and B.



A $A = \frac{1}{2}, B = 0$

B $A = 0, B = \frac{1}{2}$

C $A = -\frac{1}{2}, B = 0$

D $A = 0, B = -\frac{1}{2}$ ✓

WORKED SOLUTION & VERIFICATION

We are given the differential equation $y'' - y = \sin x$ and a proposed particular solution $y_p(x) = A\cos x + B\sin x$.

We need to find the derivatives of $y_p(x)$ and substitute them into the differential equation.

First derivative: $y_p'(x) = -A\sin x + B\cos x$.

Second derivative: $y_p''(x) = -A\cos x - B\sin x$.

Substitute these into the differential equation:

$(-A\cos x - B\sin x) - (A\cos x + B\sin x) = \sin x$

Combine like terms:

$(-A - A)\cos x + (-B - B)\sin x = \sin x$

$-2A\cos x - 2B\sin x = \sin x$

For this equation to hold for all x, the coefficients of $\cos x$ and $\sin x$ on both sides must be equal.

Comparing coefficients of $\cos x$: $-2A = 0$, which implies $A = 0$.

Comparing coefficients of $\sin x$: $-2B = 1$, which implies $B = -\frac{1}{2}$.

Therefore, the values of the constants are $A = 0$ and $B = -\frac{1}{2}$.

11 MCQ

Consider the second-order linear non-homogeneous differential equation: $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. Suppose a particular solution is found using the method of variation of parameters. Let the fundamental set of solutions for the associated homogeneous equation be $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$. If the particular solution is expressed in the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$, determine the value of $u_2'(x)$.

A $\frac{1}{4}$

B $\frac{1}{2}$ ✓

C $\frac{1}{2}\sin(2x)$

D $\frac{1}{4}\cos(2x)$

WORKED SOLUTION & VERIFICATION

The given differential equation is $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. The associated homogeneous equation is $\frac{d^2y}{dx^2} + 4y = 0$. The characteristic equation is $r^2 + 4 = 0$, which has roots $r = \pm 2i$. Thus, the fundamental set of solutions for the homogeneous equation is $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$.

We use the method of variation of parameters to find a particular solution of the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$.

The Wronskian of y_1 and y_2 is:

$W(y_1, y_2)(x) = y_1y_2' - y_1'y_2 = \cos(2x)(2\cos(2x)) - (-\sin(2x))(2\sin(2x))$

$W(y_1, y_2)(x) = 2\cos^2(2x) + 2\sin^2(2x) = 2(\cos^2(2x) + \sin^2(2x)) = 2.$

The non-homogeneous term is $g(x) = \sec(2x)$.

The formulas for $u_1'(x)$ and $u_2'(x)$ are:

$u_1'(x) = -\frac{y_2(x)g(x)}{W(x)}$

$u_2'(x) = \frac{y_1(x)g(x)}{W(x)}$

Substituting the given functions:

$u_2'(x) = \frac{\cos(2x)\cdot\sec(2x)}{2}$

$u_2'(x) = \frac{\cos(2x)}{2\cos(2x)}$

$u_2'(x) = \frac{1}{2}.$

The value of $u_2'(x)$ is $\frac{1}{2}$.

12 MCQ

The separable initial value problem $\frac{dF}{dw} = 12w$ satisfies $F(0) = 9$. Determine the solution value $F(12)$.

A 873 ✓

B 876

C 879

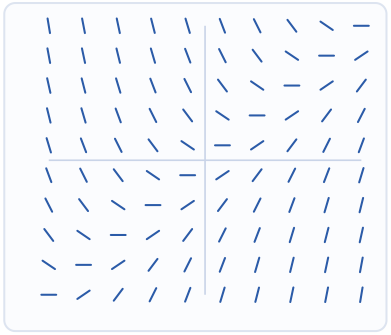
D 864

WORKED SOLUTION & VERIFICATION

Integrating gives $F(w) = 6w^2 + 9$, so $F(12) = 873$.

13 MCQ

Solve $y'' - 2y' + 5y = 0$, $y(0) = 2$, $y'(0) = 1$. Find $y(\pi/2)$.



A $2e^{\pi/2}$

B $-2e^{\pi}$

C $-e^{\pi/2}$

D $-2e^{\pi/2}$ ✓

WORKED SOLUTION & VERIFICATION

Auxiliary: $r^2 - 2r + 5 = 0 \Rightarrow r = \dfrac{2 \pm \sqrt{4-20}}{2} = 1 \pm 2i - \alpha = 1, \beta = 2$, so $y = e^x(C_1 \cos 2x + C_2 \sin 2x)$. $y(0) = C_1 = 2$; $y' = e^x[(C_1 + 2C_2)\cos 2x + (C_2 - 2C_1)\sin 2x]$, $y'(0) = C_1 + 2C_2 = 1 \Rightarrow C_2 = -\frac{1}{2}$. At $x = \pi/2$: $\cos \pi = -1, \sin \pi = 0$, so $y(\pi/2) = e^{\pi/2}(2(-1) - \frac{1}{2}(0)) = -2e^{\pi/2}$.

14 MCQ

Find a particular solution of $y'' - 4y' + 3y = 6x + 4$.

A $2x - 4$

B $2x + 2$

C $2x + 4$ ✓

D $4x + 4$

WORKED SOLUTION & VERIFICATION

Homogeneous roots are $r = 1, 3$ (from $r^2 - 4r + 3 = 0$), and $r = 0$ isn't among them, so no resonance: trial $y_p = Ax + B$ gives $y_p' = A$, $y_p'' = 0$. Substituting: $-4A + 3(Ax + B) = 3Ax + (3B - 4A) = 6x + 4$. Matching: $3A = 6 \Rightarrow A = 2$; $3B - 4A = 4 \Rightarrow 3B = 4 + 8 = 12 \Rightarrow B = 4$. So $y_p = 2x + 4$.

15 MCQ

Solve $y'' - 3y' - 4y = 0$, $y(0) = -4$, $y'(0) = -1$. Find $y(1)$.

A $-3e^4 - e^{-1}$

B $-e^4 - \frac{3}{e}$ ✓

C $\frac{3}{e} + e^4$

D $-\frac{3}{e} + e^4$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 - 3r - 4 = 0$ has roots $r = -1, 4$, so $y = C_1e^{-x} + C_2e^{4x}$. Solving $y(0) = C_1 + C_2 = -4$ and $y'(0) = -C_1 + 4C_2 = -1$ gives $C_1 = -3$, $C_2 = -1$. So $y = -3e^{-x} - e^{4x}$, and $y(1)$ follows by substitution.

16 MCQ

Obtain the value(s) satisfying $y'' - 5y' + 4y = 0$, $y(0) = 3$, $y'(0) = 6$. Find $y(1)$.

- **A** $2e + e^4$ ✓
- **B** $-2e + e^4$
- **C** $e + 2e^4$
- **D** $-e^4 - 2e$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 - 5r + 4 = 0$ has roots $r = 4, 1$, so $y = C_1 e^{4x} + C_2 e^x$. Solving $y(0) = C_1 + C_2 = 3$ and $y'(0) = 4C_1 + C_2 = 6$ gives $C_1 = 1$, $C_2 = 2$. So $y = e^{4x} + 2e^x$, and $y(1)$ follows by substitution.

17 MCQ

Solve $y'' - 2y' + y = 0$, $y(0) = -2$, $y'(0) = -5$. Find $y(1)$.

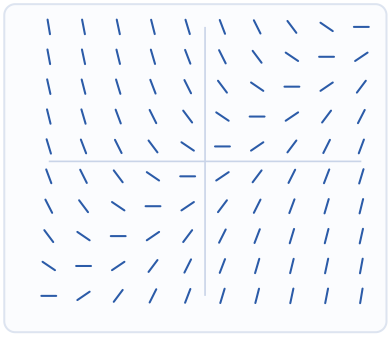
- **A** $-\frac{5}{e}$
- **B** e
- **C** $5e$
- **D** $-5e$ ✓

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 - 2r + 1 = 0 \Rightarrow (r-1)^2 = 0 \Rightarrow r = 1$ (repeated), so $y = (C_1 + C_2 x)e^x$. $y(0) = C_1 = -2$; $y'(0) = C_2 + C_1 = -5 \Rightarrow C_2 = -3$. So $y = (-2 - 3x)e^x$, and $y(1)$ follows by substitution.

18 MCQ

Solve $y'' + y = 0$, $y(0) = -3$, $y'(0) = -3$. Find $y(\pi)$.



- **A** $\frac{3}{2}$
- **B** 3 ✓
- **C** -3
- **D** 6

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 + 1 = 0$ has roots $r = 0 \pm 1i$, so $y = C_1 \cos x + C_2 \sin x$. $y(0) = C_1 = -3$; $y'(0) = C_2 = -3 \Rightarrow C_2 = -3$. Evaluating at $x = \pi$ gives the result.

19 MCQ

Solve $y'' + 9y = 0$, $y(0) = 3$, $y'(0) = 9$. Find $y(\frac{\pi}{6})$.

A -3

B 3 ✓

C 6

D $\frac{3}{2}$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 + 9 = 0$ has roots $r = 0 \pm 3i$, so $y = C_1 \cos 3x + C_2 \sin 3x$. $y(0) = C_1 = 3$; $y'(0) = 3C_2 = 9 \Rightarrow C_2 = 3$. Evaluating at $x = \frac{\pi}{6}$ (where $3x = \frac{\pi}{2}$) gives the result.

20 MCQ

Find the root(s) satisfying $y'' - 7y' + 12y = 0$, $y(0) = 4$, $y'(0) = 13$. Find $y(2)$.

A $3e^6 + e^8$ ✓

B $-3e^6 + e^8$

C $e^6 + 3e^8$

D $-e^8 - 3e^6$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 - 7r + 12 = 0$ has roots $r = 4, 3$, so $y = C_1e^{4x} + C_2e^{3x}$. Solving $y(0) = C_1 + C_2 = 4$ and $y'(0) = 4C_1 + 3C_2 = 13$ gives $C_1 = 1$, $C_2 = 3$. So $y = e^{4x} + 3e^{3x}$, and $y(2)$ follows by substitution.

21 MCQ

Obtain the value(s) satisfying $y'' - 4y' + 3y = 0$, $y(0) = 0$, $y'(0) = 8$. Find $y(2)$.

A $-4e^6 + 4e^2$

B $-4e^2 + 4e^6$ ✓

C $-8e^2 + 8e^6$

D $-4e^6 - 4e^2$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 - 4r + 3 = 0$ has roots $r = 1, 3$, so $y = C_1e^x + C_2e^{3x}$. Solving $y(0) = C_1 + C_2 = 0$ and $y'(0) = C_1 + 3C_2 = 8$ gives $C_1 = -4$, $C_2 = 4$. So $y = -4e^x + 4e^{3x}$, and $y(2)$ follows by substitution.

22 MCQ

Calculate the solution set for $y'' - 4y' + 4y = 0$, $y(0) = -4$, $y'(0) = -11$. Find $y(2)$.

A $2e^4$

B $-11e^4$

C $10e^4$

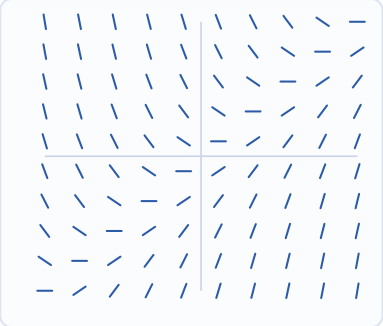
D $-10e^4$ ✓

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 - 4r + 4 = 0 \Rightarrow (r-2)^2 = 0 \Rightarrow r = 2$ (repeated), so $y = (C_1 + C_2x)e^{2x}$. $y(0) = C_1 = -4$; $y'(0) = C_2 + 2C_1 = -11 \Rightarrow C_2 = -3$. So $y = (-4 - 3x)e^{2x}$, and $y(2)$ follows by substitution.

23 MCQ

Obtain the value(s) satisfying $y'' + 8y' + 16y = 0$, $y(0) = -2$, $y'(0) = 5$. Find $y(2)$.



A $\frac{4}{e^8}$

B $-\frac{7}{e^8}$

C $-\frac{8}{e^8}$ ✓

D $\frac{8}{e^8}$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 + 8r + 16 = 0 \Rightarrow (r+4)^2 = 0 \Rightarrow r = -4$ (repeated), so $y = (C_1 + C_2 x)e^{-4x}$. $y(0) = C_1 = -2$; $y'(0) = C_2 - 4C_1 = 5 \Rightarrow C_2 = -3$. So $y = (-2 - 3x)e^{-4x}$, and $y(2)$ follows by substitution.

24 MCQ

Calculate the solution set for $y'' + 4y' + 5y = 0$, $y(0) = 2$, $y'(0) = -2$. Find $y(\frac{\pi}{4})$.

A 0

B $-\frac{2\sqrt{2}}{e^{\frac{\pi}{2}}}$

C $\frac{2\sqrt{2}}{e^{\frac{\pi}{2}}}$ ✓

D $2\sqrt{2}e^{\frac{\pi}{2}}$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 + 4r + 5 = 0$ has roots $r = -2 \pm 1i$, so $y = e^{-2x}(C_1 \cos x + C_2 \sin x)$. $y(0) = C_1 = 2$; $y'(0) = -2C_1 + C_2 = -2 \Rightarrow C_2 = 2$. Evaluating at $x = \frac{\pi}{4}$ gives the result.

25 MCQ

Calculate the solution set for $y'' - 4y' - 45y = 0$, $y(0) = -7$, $y'(0) = -7$. Find $y(1)$.

A $-3e^9 + \frac{4}{e^5}$

B $-4e^9 - \frac{3}{e^5}$

C $-3e^9 - \frac{4}{e^5}$ ✓

D $\frac{4}{e^5} + 3e^9$

WORKED SOLUTION & VERIFICATION

Auxiliary equation $r^2 - 4r - 45 = 0$ has roots $r = 9, -5$, so $y = C_1 e^{9x} + C_2 e^{-5x}$. Solving $y(0) = C_1 + C_2 = -7$ and $y'(0) = 9C_1 - 5C_2 = -7$ gives $C_1 = -3$, $C_2 = -4$. So $y = -3e^{9x} - 4e^{-5x}$, and $y(1)$ follows by substitution.